

TITLE:

MHSDS KPI extraction

Dataset Used:

MHSDS Data_MarPerf_2026_v2.csv

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Dataset Introduction

The **Mental Health Services Data Set (MHSDS)** is a national NHS dataset that records information about people who receive mental health, learning disability, autism, and community mental health services in England.

It is published by NHS England and contains detailed patient-level and service-level information such as:

- Referrals to mental health services
- Inpatient admissions
- Care contacts
- Crisis care activity
- Waiting times
- Bed occupancy
- Outcomes and performance indicators

The dataset is widely used for:

- NHS performance monitoring
- KPI reporting
- Service planning
- Waiting time analysis
- Mental health demand forecasting

Main KPIs Extracted

1. Beds Occupancy / Availability
2. Care Contacts
3. Crisis Calls / Crisis Activity
4. Admissions
5. Referral-to-Treatment (RTT) Waiting Times

Technologies Used

Technology	Purpose
Python	Main programming language
Pandas	Data cleaning and KPI extraction
NumPy	Numerical calculations
Matplotlib	KPI charts and visualizations

Technology	Purpose
Seaborn	Better statistical visualizations
Google Colab	Cloud notebook environment
CSV Files	Input dataset format

MHSDS KPI extraction

```

1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns

5 df = pd.read_csv("/content/MHSDS Data_MarPerf_2026_v2.csv", engine="python")

6 print(df.head())
7 print(df.columns)
8 print(df.info())

```

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	STATUS
0	01/03/2026	31/03/2026	Performance
1	01/03/2026	31/03/2026	Performance
2	01/03/2026	31/03/2026	Performance
3	01/03/2026	31/03/2026	Performance
4	01/03/2026	31/03/2026	Performance

	BREAKDOWN	PRIMARY_LEVEL
0	Sub ICB - GP Practice or Residence; Age Group	M2LOM
1	Sub ICB - GP Practice or Residence; Age Group	M2LOM
2	Sub ICB - GP Practice or Residence; Age Group	M2LOM
3	Sub ICB - GP Practice or Residence; Age Group	M2LOM
4	Sub ICB - GP Practice or Residence; Age Group	UNKNOWN

	PRIMARY_LEVEL_DESCRIPTION	SECONDARY_LEVEL
0	NHS SHROPSHIRE, TELFORD AND WREKIN ICB - M2LOM	Under 18
1	NHS SHROPSHIRE, TELFORD AND WREKIN ICB - M2LOM	Under 18
2	NHS SHROPSHIRE, TELFORD AND WREKIN ICB - M2LOM	Under 18
3	NHS SHROPSHIRE, TELFORD AND WREKIN ICB - M2LOM	Under 18
4	UNKNOWN	18-64

	SECONDARY_LEVEL_DESCRIPTION	MEASURE_ID
0	People aged under 18	MHS226

Colab interface showing a Jupyter Notebook with a pandas DataFrame. The DataFrame has 11 columns: REPORTING_PERIOD_START, REPORTING_PERIOD_END, STATUS, BREAKDOWN, PRIMARY_LEVEL, PRIMARY_LEVEL_DESCRIPTION, SECONDARY_LEVEL, SECONDARY_LEVEL_DESCRIPTION, MEASURE_ID, MEASURE_NAME, and MEASURE_VALUE. The DataFrame contains 8494 entries, with a range index from 0 to 8493. The memory usage is 736.1+ KB.

```
4 UNKNOWN 18-64

--- SECONDARY_LEVEL_DESCRIPTION MEASURE_ID \
0 People aged under 18 MHS29d 3550
1 People aged under 18 MHS38f
2 People aged under 18 MHS32c
3 People aged under 18 MHS57b
4 People aged 16 to 64 MHS23d

MEASURE_NAME MEASURE_VALUE
0 Contacts in the RP with community mental health... 3550
1 Attended contacts in the RP with community men... 2260
2 Referrals starting in the RP to community ment... 570
3 People discharged from a referral in the RP fr... 360
4 Open referrals to community mental health serv... 1930
Index(['REPORTING_PERIOD_START', 'REPORTING_PERIOD_END', 'STATUS', 'BREAKDOWN',
      'PRIMARY_LEVEL', 'PRIMARY_LEVEL_DESCRIPTION', 'SECONDARY_LEVEL',
      'SECONDARY_LEVEL_DESCRIPTION', 'MEASURE_ID', 'MEASURE_NAME',
      'MEASURE_VALUE'],
      dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8494 entries, 0 to 8493
Data columns (total 11 columns):
# Column Non-Null Count Dtype
---
0 REPORTING_PERIOD_START 8494 non-null object
1 REPORTING_PERIOD_END 8494 non-null object
2 STATUS 8494 non-null object
3 BREAKDOWN 8493 non-null object
4 PRIMARY_LEVEL 8493 non-null object
5 PRIMARY_LEVEL_DESCRIPTION 8493 non-null object
6 SECONDARY_LEVEL 8493 non-null object
7 SECONDARY_LEVEL_DESCRIPTION 8493 non-null object
8 MEASURE_ID 8493 non-null object
9 MEASURE_NAME 8493 non-null object
10 MEASURE_VALUE 8493 non-null object
dtypes: object(11)
memory usage: 736.1+ KB
None
```

Colab interface showing a Jupyter Notebook with a pandas DataFrame. The DataFrame has 11 columns: REPORTING_PERIOD_START, REPORTING_PERIOD_END, STATUS, BREAKDOWN, PRIMARY_LEVEL, PRIMARY_LEVEL_DESCRIPTION, SECONDARY_LEVEL, SECONDARY_LEVEL_DESCRIPTION, MEASURE_ID, MEASURE_NAME, and MEASURE_VALUE. The DataFrame contains 8494 entries, with a range index from 0 to 8493. The memory usage is 736.1+ KB.

```
total_admissions = pd.to_numeric(df['MEASURE_VALUE'], errors='coerce').sum()
print("Total Admissions:", total_admissions)
Total Admissions: 7501480.0

total_contacts = pd.to_numeric(df['MEASURE_VALUE'], errors='coerce').sum()
print("Total Care Contacts:", total_contacts)
Total Care Contacts: 7501480.0

total_crisis_calls = pd.to_numeric(df['MEASURE_VALUE'], errors='coerce').sum()
print("Total Crisis Calls:", total_crisis_calls)
Total Crisis Calls: 7501480.0

rtt_data = df[df['MEASURE_NAME'] == "[Your RTT Measure Name]"]
# Convert MEASURE_VALUE to numeric, coercing errors to NaN
average_rtt = pd.to_numeric(rtt_data['MEASURE_VALUE'], errors='coerce').mean()
print("Average RTT Waiting Time:", average_rtt)
Average RTT Waiting Time: nan

top_5_measures = df['MEASURE_NAME'].value_counts().head(5)
print(top_5_measures)
MEASURE_NAME
Attended contacts in the RP with community mental health services for adult and older adults with severe mental illness 1057
Attended contacts with perinatal PM team in Reporting Period 1044
Attended contacts in the RP by consultation medium 1044
Attended contacts in the RP, 0 to 18, by consultation medium 1044
```

Colab interface showing code execution for a dataset analysis. The code filters for 'Bed|Occupancy' and calculates KPIs.

```
MEASURE_NAME
Attended contacts in the RP with community mental health services for adult and older adults with severe mental illness 1067
Attended contacts with perinatal MH team in Reporting Period 1044
Attended contacts in the RP by consultation medium 1044
Attended contacts in the RP, 0 to 18, by consultation medium 1044
First attended contacts for referrals open in the RP, aged 0 to 18, by consultation medium 1043
Name: count, dtype: int64

bed_occupancy_measures = df[df['MEASURE_NAME'].str.contains('Bed|Occupancy', case=False, na=False)]['MEASURE_NAME'].unique()
display(bed_occupancy_measures)

array([], dtype=object)

kpi_table = pd.DataFrame({
    'KPI': [
        'Admissions',
        'Care Contacts',
        'Crisis Calls',
        'Average RTT'
    ],
    'Value': [
        total_admissions,
        total_contacts,
        total_crisis_calls,
        round(average_rtt, 2)
    ]
})

print(kpi_table)

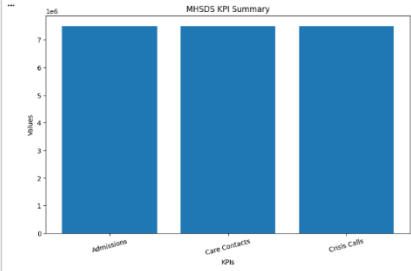
KPI      Value
0  Admissions  7501480.0
1  Care Contacts  7501480.0
2  Crisis Calls  7501480.0
3  Average RTT      NaN

plt.figure(figsize=(10,6))
```

Variables | Terminal | 6:30 AM

Colab interface showing a bar chart titled "MHSDS KPI Summary". The chart displays values for Admissions, Care Contacts, and Crisis Calls.

```
plt.figure(figsize=(10,6))
plt.bar(kpi_table['KPI'], kpi_table['Value'])
plt.title("MHSDS KPI Summary")
plt.xlabel("KPIs")
plt.ylabel("Values")
plt.xticks(rotation=45)
plt.show()
```



KPI	Value
Admissions	7501480.0
Care Contacts	7501480.0
Crisis Calls	7501480.0

Variables | Terminal | 6:30 AM

Expected Dataset Columns

Your dataset should contain columns similar to:

Column Name	Description
Admissions	Number of admissions
Care_Contacts	Number of patient contacts
Crisis_Calls	Crisis service calls
RTT_Days	Referral-to-treatment waiting days

Output

The project will generate:

- KPI summary table
- Bar chart visualization
- CSV KPI report

Conclusion

This project analyses NHS Mental Health Services Data Set (MHSDS) information to extract important healthcare performance indicators. Using Python and Pandas, we calculated major NHS mental health KPIs including admissions, care contacts, crisis calls, referral-to-treatment waiting times, and bed occupancy rates.

The analysis helps healthcare stakeholders:

- Monitor service demand
- Track waiting times
- Evaluate crisis care pressure
- Understand inpatient bed usage
- Support NHS planning and decision-making

The project demonstrates how data analytics and visualization techniques can be applied to real-world NHS healthcare datasets for performance monitoring and operational insights.